

Week 3 - Exercises on Bayesian Perception

The brain needs to integrate multiple sensory streams to obtain a representation about of the world. As we saw in today's lecture, a typical example is the integration of haptic and visual feedback. Furthermore, inference is also important for motor control and learning. Bayesian statistics offers a theoretical framework to interpret how humans and animals solve the sensory integration problem. Overall, the goal of today's exercise is to:

- Learn to derive step-by-step the rule used by a decision making system to fuse different sensory information, according to the Bayesian interpretation.
- Compare the Bayesian and the maximum likelihood interpretation of sensory integration.
- Find the minimum variance weighted average of the sensory inputs.

Exercise 1.1 - Integration of two sensory channels

Let us consider the two Gaussian random variables t and v , corresponding to the (noisy) estimation of the width w of an object through tactile and visual feedback. We assume that the two variables are independent, and that they are distributed according to the following conditional density function:

$$\begin{aligned} p(t|w) &= \frac{1}{\sqrt{2\pi\sigma_t^2}} \exp\left(-\frac{(w-t)^2}{2\sigma_t^2}\right) \\ p(v|w) &= \frac{1}{\sqrt{2\pi\sigma_v^2}} \exp\left(-\frac{(w-v)^2}{2\sigma_v^2}\right) \end{aligned} \quad (1)$$

Show that, assuming a flat prior distribution $p(w)$, the posterior is proportional to the product of the likelihoods, i.e.,

$$p(w|t, v) \propto p(t|w)p(v|w) \quad (2)$$

Exercise 1.2

Re-write the analytical expression of $p(t|w)p(v|w)$ to show that it is the density function of a Gaussian distribution of variable w (up to a normalization coefficient independent of w).

Exercise 1.3

Given the density function derived in Exercise 1.2, extract the mean μ_w and the variance σ_w^2 of w as a function of t , v , σ_t and σ_v , and show that μ_w is the weighted average of t and v , with weights proportional to the inverse of their variance.

Exercise 1.4

Show that the variance of W is smaller than both the variance of V and of the variance of T , for any value of $\sigma_t > 0$ and $\sigma_v > 0$.

Exercise 2.1 - Extension to multiple sensory channels

Consider a new estimator u of w , through another sensory input (or sensor), independent of v and t , and also following a Gaussian distribution:

$$p(u|w) = \frac{1}{\sqrt{2\pi\sigma_u^2}} \exp\left(-\frac{(w-u)^2}{2\sigma_u^2}\right) \quad (3)$$

Write the new expression of μ_w and σ_w^2 .

Exercise 2.2

Provide the expression of μ_w and σ_w^2 given an arbitrary number N of independent Gaussian estimators $t_1, \dots, t_N$, normally distributed with mean w and variance σ_i , $i = 1, \dots, N$.

Exercise 3.1 - Maximum likelihood approach

Let us now consider the problem of multisensory integration from a different point of view. Instead of the Bayesian approach, we want to estimate w through maximum likelihood. Consider the conditional distributions $p(v|w)$ and $p(t|w)$ as before and compute the maximum likelihood estimator for μ_w . Derive the variance of such estimator and show that you retrieve the same result as in Exercise 1.3. Can the variance of an unbiased estimator of μ_w be lower than the value you have found?

Exercise 3.2

In the derivation through maximum likelihood, we have implicitly assumed that all the information about w is provided by v and t . How was this hypothesis included in the Bayesian multisensory integration?

Exercise 4.1 - Minimum variance weighted average

We consider once again the problem of sensory integration from a different point of view. Let us consider a weighted sum of noisy "measurements" of w , defined as $W = \sum_{i=1}^N \alpha_i T_i$, in which the index i indicates a sensory channel. The measurements t_i are independent random variables for T_i with mean w and variance σ_i^2 , $i = 1, \dots, N$. First, show that the necessary and sufficient condition for W to be unbiased is that $\sum_i \alpha_i = 1$. Second, use the technique of the Lagrange multiplier to show that the coefficients α_i are the ones found in Exercise 2.2.